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INSERTION DEVICE FOR A BIOSENSOR

Abstract

An insertion device includes a cover assembly, an implant module, a patch base, a sensor assembly, and a peel-off component. The cover assembly includes an outer shell and a bottom cover. The outer shell defines an accommodating space. The implant module is disposed in the accommodating space and is operable for implanting and extracting a needle. The patch base includes a base body, an adhesive pad that is connected fixedly to the base body, and an outer release layer that is attached detachably to the adhesive pad. The sensor assembly is connected to the base body when the needle is implanted by the implant module. The peel-off component is connected to the bottom cover and the outer release layer. The peel-off component removes the outer release layer from the adhesive pad when the bottom cover is separated from the outer shell.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/561,355, filed on Mar. 5, 2024, the entire disclosure of which is incorporated by reference herein.

FIELD

[0002] The disclosure relates to biological detection system and method that are applied to an organism, and more particularly to an insertion device for installing a biological detection device onto the skin of the organism and for implanting a biosensor under the skin of the organism.

BACKGROUND

[0003] Conventional self-testing of blood glucose is conducted through drawing blood from capillaries of the fingertip, disposing the blood on a blood glucose test strip, and using a machine to read the blood glucose level. As long as the test is conducted correctly, the blood glucose level obtained will be very accurate. When the obtained blood glucose level is too high or too low, medical care can be given as soon as possible. Some people may need to test their blood glucose regularly, especially patients who have poor blood glucose control, are under insulin treatment, or have large blood glucose fluctuations.

[0004] Another device can also achieve self-testing of blood glucose, which is a continuous glucose monitor (CGM). A sensor needle is disposed underneath the patient's skin to continuously monitor glucose concentration in tissue fluid (glucose in blood diffuses into tissue fluid to enter cells). The sensor estimates blood glucose levels via calculation, and can provide immediate blood sugar level values. The sensor periodically records blood glucose levels, and shows a trend chart of the changes in blood glucose, so as to send out a warning when the blood glucose level is too high or too low.

[0005] Current studies show that, for patients who, with type 1 or type 2 diabetes, need insulin injections, using the CGM for testing, in comparison to testing blood glucose through drawing blood from the fingertip, lowers glycohemoglobin by 0.6% and decreases a total amount of time during which low blood glucose occurs each day.

[0006] Since the CGM is worn by a user for long periods of time, the trend of reducing the size of the CGM is inevitable. A structure of the CGM includes: (a) a sensor for detecting a biosignal that corresponds to glucose concentration in the human body, (b) a receiver for receiving and transmitting the biosignal, and (c) an insertion device for attaching the sensor to the receiver, and for inserting the sensor under the user's skin.

[0007] To achieve safe and accurate implantation of the sensor under the user's skin, and to have the biosignal sensed by the sensor transmitted to corresponding receiving equipment of the receiver to allow the user to obtain the state of their blood glucose at any time, the applicant has filed two applications, U.S. Patent Application Publications No. 20210030960A1 and No. 20210030344A1. However, the applicant believes that the functions of the insertion devices disclosed in the abovementioned applications can be further improved to provide more convenient use and a shorter period of sensor implementation time. The applicant hopes the improved insertion device can make assembly easier in manufacturing and production processes to increase production yield and speed.

SUMMARY

[0008] Therefore, an object of the disclosure is to provide an insertion device that can alleviate at least one of the drawbacks of the prior art.

[0009] According to the disclosure, the insertion device is adapted to be installed in a biosensor. The insertion device includes a cover assembly, an implant module, a patch base, a sensor assembly, and a peel-off component. The cover assembly includes an outer shell and a bottom cover that is removably coupled to the outer shell. The outer shell defines an accommodating space. The implant module is disposed in the accommodating space and is operable for implanting and extracting a needle. The patch base is positioned removably relative to the implant module, and includes a base body, an adhesive pad, and an outer release layer. The adhesive pad is connected fixedly to the base body. The outer release layer is attached detachably to the adhesive pad. The sensor assembly is connected to the base body when the needle is implanted by the implant module. The peel-off component is connected to the bottom cover and the outer release layer. The peel-off component removes the outer release layer from the adhesive pad when the bottom cover is separated from the outer shell.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Other features and advantages of the disclosure will become apparent in the following detailed description of the embodiment(s) with reference to the accompanying drawings. It is noted that various features may not be drawn to scale.

[0011] FIG. 1 is a sectional perspective view illustrating an embodiment of an insertion device according to the disclosure.

[0012] FIG. 2 is a sectional perspective view taken along line II-II in FIG. 1.

[0013] FIG. 3 is a sectional view taken along line III-III in FIG. 1, illustrating a configuration of a plurality of positioner columns.

[0014] FIG. 4 is a partially exploded perspective view of the embodiment.

[0015] FIG. 5 is a fragmentary exploded perspective view of the embodiment.

[0016] FIG. 6 is a perspective view of an inner lining member of the embodiment.

[0017] FIG. 7 is a perspective view of a main body member of the embodiment.

[0018] FIG. 8 is an exploded schematic perspective view of an insertion seat, a needle extraction member, and the positioner columns of the embodiment.

[0019] FIG. 9 is a fragmentary sectional view of the insertion seat.

[0020] FIG. 10 is a top view of the needle extraction member.

[0021] FIG. 11 is a bottom view of a peel-off component and a patch base of the embodiment.

[0022] FIG. 12 is an exploded perspective view of the peel-off component and the patch base.

[0023] FIG. 13 is a sectional view of the embodiment in an unfired state, illustrating a bottom cover of the embodiment being attached to an outer shell of the embodiment.

[0024] FIG. 14 is a sectional view taken along line XIV-XIV in FIG. 13.

[0025] FIG. 15 is a schematic sectional view of the embodiment in a pre-firing state, illustrating the bottom cover being separated from the outer shell.

[0026] FIG. 16 is a schematic sectional view, illustrating the outer shell being pressed after the bottom cover has been removed, and the embodiment being in a firing state.

[0027] FIG. 17 is a sectional view taken along line XVII-XVII in FIG. 16.

[0028] FIG. 18 is a schematic sectional view of the embodiment in the firing state, illustrating a firing restriction structure being unlocked.

[0029] FIG. 19 is a sectional view taken along line XIX-XIX in FIG. 18.

[0030] FIG. 20 is a fragmentary schematic sectional view of the embodiment, illustrating a needle being fired.

[0031] FIG. 21 is a sectional view taken along line XXI-XXI in FIG. 20.

[0032] FIG. 22 is a fragmentary schematic sectional view of the embodiment in a post-fired state.

[0033] FIG. **23** is a sectional view taken along line XXIII-XXIII in FIG. **22**.

[0034] FIG. **24** is a fragmentary schematic sectional view of the embodiment, illustrating the needle being extracted.

[0035] FIG. **25** is a sectional view taken along line XXV-XXV in FIG. **24**.

[0036] FIG. **26** is a fragmentary schematic sectional view of the embodiment after the patch base, a sensor assembly, and the main body member have been removed.

DETAILED DESCRIPTION

[0037] It should be noted herein that for clarity of description, spatially relative terms such as “top,” “bottom,” “upper,” “lower,” “on,” “above,” “over,” “downwardly,” “upwardly” and the like may be used throughout the disclosure while making reference to the features as illustrated in the drawings. The features may be oriented differently (e.g., rotated 90 degrees or at other orientations) and the spatially relative terms used herein may be interpreted accordingly.

[0038] Referring to FIGS. **1** to **5**, an embodiment of an insertion device for a biosensor according to the disclosure includes a cover assembly **1**, an implant module **2**, two fixture members **3**, a patch base **4**, a sensor assembly **5**, a peel-off component **6** (see FIG. **12**), and a desiccant **7**.

[0039] The cover assembly **1** includes an outer shell **10**, a top cap **20** fixed in the outer shell **10**, an inner lining member **30** disposed in the outer shell **10** and at a side of the top cap **20**, and a bottom cover **40** airtightly and removably coupled to the outer shell **10**. The outer shell **10** defines an accommodating space **11**. The top cap **20** and the outer shell **10** cooperatively define a receiving space **21** that takes up a portion of the accommodating space **11**, and that is isolated from the remaining portions of the accommodating space **11**. The top cap **20** has an open hole **22** for disposing the desiccant **7** (which may be, for example, a desiccant pack, a desiccant tablet, or desiccant beads) into the receiving space **21**. In this embodiment, the desiccant **7** is smaller than the open hole **22**. During assembly of the insertion device, after the desiccant **7** is disposed in the receiving space **21**, the open hole **22** is blocked (e.g., by an additional component) but not sealed, so the desiccant **7** is still able to keep the contents in the insertion device dry. In other embodiments, if the desiccant **7** is a desiccant package that is bigger than the open hole **22**, there will be no need to block the open hole **22** by an additional component.

[0040] Referring to FIG. **6**, the inner lining member **30** is hollow and ring-shaped, and has an inner peripheral surface **31**, an outer peripheral surface **32** opposite to the inner peripheral surface **31**, a plurality of firing portions **33** disposed on the inner peripheral surface **31**, a pair of latch portions **34** being elastic and extending downwards from the outer peripheral surface **32**, a pair of push members **35** disposed at a bottom end of the inner lining member **30** and extending downwardly, and a fourth alignment label **36** disposed on the outer peripheral surface **32**. The fourth alignment label **36** is a triangular label.

[0041] The bottom cover **40**, when being coupled to the outer shell **10**, faces an opening of the accommodating space **11**. The bottom cover **40** includes a bottom plate portion **41**.

[0042] The implant module **2** is disposed in the accommodating space **11** of the cover assembly **1**. The implant module **2** includes a main body unit **50**, a positioner unit **60**, an implant seat **70**, a first resilient member **81**, a needle extraction member **90**, a second resilient member **82** and a needle implant member **100**.

[0043] The main body unit **50** is connected to the cover assembly **1**, and is slidably disposed in the inner lining member **30**. The main body unit **50** includes a main body member **51**, and a main body cap **52** that is removably connected to the main body member **51**. The main body member **51** and the main body cap **52** cooperatively define a movement space **53**. Referring to FIG. **7**, the main body member **51** has a bottom wall **511**, an annular wall **512** connected to the bottom wall **511**, a plurality of engagement portions **513** disposed on the annular wall **512**, a plurality of buckle stopper portions **514** formed on the annular wall **512**, a pair of slide grooves **515** extending through the annular wall **512**, recessing into the bottom wall **511**, and spatially communicating with the movement space **53**, a pair of lock portions **516** extending from the annular wall **512** and

respectively corresponding in position to the latch portions **34**, and a first alignment label **517** disposed on the annular wall **512**. The bottom wall **511** has four lower positioning holes **518**. An angle is formed between every two adjacent lower positioning holes **518**, and there are four angles ($\theta_{\text{sub.1}}$, $\theta_{\text{sub.2}}$, $\theta_{\text{sub.3}}$, $\theta_{\text{sub.4}}$). The angles ($\theta_{\text{sub.1}}$, $\theta_{\text{sub.2}}$, $\theta_{\text{sub.3}}$, $\theta_{\text{sub.4}}$) are not symmetrically arranged. The first alignment label **517** corresponds in position to the fourth alignment label **36**. The first alignment label **517** is a triangular label. Referring to FIG. 5, the main body cap **52** has a plurality of connection portions **521** that are respectively engaged with the engagement portions **513**, two limiter members **522** (see FIG. 2) that are elongated, a second alignment label **523** that corresponds in position to the first alignment label **517**, and four upper positioning holes **524**. The second alignment label **523** is a triangular label. An angle is formed between every two adjacent upper positioning holes **524**, and there are four angles. The angles formed between every two adjacent upper positioning holes **524** are respectively identical to the angles ($\theta_{\text{sub.1}}$, $\theta_{\text{sub.2}}$, $\theta_{\text{sub.3}}$, $\theta_{\text{sub.4}}$) formed between every two adjacent lower positioning holes **518**, and are not symmetrically arranged.

[0044] Referring to FIG. 8, the positioner unit **60** is disposed in the movement space **53** (see FIG. 5), and includes four positioner columns **61** that interconnect the main body member **51** (see FIG. 5) and the main body cap **52** (see FIG. 5). The positioner columns **61** extend in an axial direction. Referring to FIG. 3, each of the positioner columns **61** has an end disposed in a respective one of the lower positioning holes **518**, and an axially-opposite end disposed in a respective one of the upper positioning holes **524**. An angle is formed between every two adjacent positioner columns **61**, and there are four angles. The angles formed between every two adjacent positioner columns **61** are respectively identical to the angles ($\theta_{\text{sub.1}}$, $\theta_{\text{sub.2}}$, $\theta_{\text{sub.3}}$, $\theta_{\text{sub.4}}$) formed between every two lower positioning holes **518**, and are not symmetrically arranged. That is to say, referring to FIG. 7, any one of the angles ($\theta_{\text{sub.1}}$, $\theta_{\text{sub.2}}$, $\theta_{\text{sub.3}}$, $\theta_{\text{sub.4}}$) is not equal to an opposite one of the angles ($\theta_{\text{sub.3}}$, $\theta_{\text{sub.4}}$, $\theta_{\text{sub.1}}$, $\theta_{\text{sub.2}}$). The positioner columns **61** are cylinders, but may be changed to other columns with different cross-section shapes as needed in other embodiments.

[0045] Referring to FIGS. 5 and 8, the implant seat **70** is removably disposed in the main body unit **50**. The positioner columns **61** extend through the implant seat **70** so that the implant seat **70** is movable along the positioner columns **61** in the movement space **53**. The implant seat **70** includes a flat plate member **71**, an outer barrel member **72** connected transversely to the flat plate member **71**, an inner barrel member **73** connected transversely to the flat plate member **71** and disposed in the outer barrel member **72**, a plurality of guide tubes **74** connected to the inner barrel member **73**, a limiter module **75** connected to the inner barrel member **73**, and positioning the needle extraction member **90** relative to the implant seat **70**, a plurality of buckle portions **76** respectively engaged with the buckle stopper portions **514** of the main body member **51** and respectively driven by the firing portions **33** to be disengaged from the buckle stopper portions **514**, four first guide holes **77**, a positioner member **78** extending downwardly from the flat plate member **71** (see FIG. 3), and a fifth alignment label **79** (see FIG. 5) disposed on the outer barrel member **72** and corresponding in position to the first alignment label **517**. The first guide holes **77** respectively extend through the guide tubes **74**. The positioner columns **61** extend respectively through the four first guide holes **77**. An angle is formed between every two adjacent first guide holes **77**, and there are four angles. The angles formed between every two adjacent first guide holes **77** are respectively identical to the angles ($\theta_{\text{sub.1}}$, $\theta_{\text{sub.2}}$, $\theta_{\text{sub.3}}$, $\theta_{\text{sub.4}}$) formed between every two adjacent lower positioning holes **518**, and are not symmetrically arranged (see FIG. 8, see FIG. 7 for the angles ($\theta_{\text{sub.1}}$, $\theta_{\text{sub.2}}$, $\theta_{\text{sub.3}}$, $\theta_{\text{sub.4}}$). Referring further to FIG. 9, each of the guide tubes **74** has a bottom portion **741** connected to the flat plate member **71**, a top portion **742** opposite to the bottom portion **741** in the axial direction in which the positioner columns **61** extend, and a hollow portion **743** disposed between the bottom portion **741** and the top portion **742**, and spatially communicating with a respective one of the first guide holes **77**. The limiter module **75** includes a pair of limiter components **751** that are hooked in shape. Each of the limiter components **751** and the inner barrel

member **73** cooperatively define a limiter slot **752**. The limiter members **522** are respectively slotted in the limiter slots **752**. The limiter module **75** restricts a position of the needle extraction member **90**, so that the implant seat **70**, the main body cap **52** and the needle extraction member **90** form a needle extraction limit structure (B). The buckle portions **76** are connected to the outer barrel member **72**. The buckle portions **76** cooperate with the buckle stopper portions **514**; that is to say, the buckle portions **76** engage the buckle stopper portions **514**, so that the implant seat **70** and the main body member **51** form a needle implantation limit structure (A). The positioner member **78** has two link portions **781** (only one positioner member **78** is shown in FIG. **3**) that are columns. [0046] The first resilient member **81** has an end that abuts against the main body cap **52**, and another end that abuts against the implant seat **70**. The first resilient member **81** is a compression spring.

[0047] The needle extraction member **90** is separable from and positioned relative to the implant seat **70**, and movement thereof is guided by the positioner unit **60**. Specifically, the needle extraction member **90** has four second guide holes **91**. The positioner columns **61** respectively extend through the second guide holes **91**, so that the needle extraction seat **90** is slidable along the positioner columns **61**. The needle extraction member **90** includes a third alignment label **92** that corresponds in position to the first alignment label **517**. An angle is formed between every two adjacent second guide holes **91**, and there are four angles. The angles formed between every two adjacent second guide holes **91** are respectively identical to the angles ($\theta_{sub.1}$, $\theta_{sub.2}$, $\theta_{sub.3}$, $\theta_{sub.4}$) formed between every two adjacent lower positioning holes **518**, and are not symmetrically arranged (see FIG. **10**).

[0048] The second resilient member **82** is disposed between and resiliently abutting against the implant seat **70** and the needle extraction member **90**. The second resilient member **82** is a compression spring.

[0049] The needle implant member **100** includes a needle implant member body **110**, and a needle **120** connected to the needle implant member body **110** for carrying a sensor **502**.

[0050] Referring to FIG. **5**, the fixture members **3** are respectively and slidably disposed in the slide grooves **515** of the main body member **51**. Each of the fixture members **3** has an abutment portion **301**, a support portion **302** opposite to the abutment portion **301**, a first hook **303** disposed between the abutment portion **301** and the support portion **302**, and a driver portion **304** disposed between the abutment portion **301** and the support portion **302** and having a driver surface **305** that is operable to be pushed by a respective one of the push members **35**. When the bottom cover **40** is coupled to the outer shell **10**, the bottom cover **40** blocks the fixture members **3**, which positions the fixture members **3** relative to the main body member **51** (i.e., the fixture members **3** are not movable relative to the main body member **51**).

[0051] Referring to FIGS. **2**, **11** and **12**, the patch base **4** is removable from and positioned relative to the main body unit **50**. The patch base **4** includes a base body **401**, a bonding layer **420**, an adhesive pad **402** connected fixedly to the base body **401**, and an outer release layer **403**. The bonding layer **420** is disposed between the base body **401** and the adhesive pad **402**. The base body **401** has a sensor positioning groove **400**, a periphery **404**, and two engagement grooves **405** recessing into a bottom side of the periphery **404**. The periphery **404** has two first sides **406**, and two second sides **407** interconnecting the first sides **406**. A length of each of the first sides **406** is equal to or shorter than a length of each of the second sides **407**. In this embodiment, the outer release layer **403** is attached detachably to the adhesive pad **402**. The base body **401** of the patch base **4** is made of a material harder than a material of the adhesive pad **402**. The first hooks **303** of the fixture members **3** are respectively and removably engaged with the engagement grooves **405**. Referring to FIGS. **11** and **12**, the adhesive pad **402** has a first notch **407'** corresponding in position to one of the first sides **406**. The outer release layer **403** has a second notch **408** corresponding in position to the first notch **407'**. A material of the base body **401** may be polycarbonate (PC), acrylonitrile butadiene styrene (ABS), a blend of polycarbonate/acrylonitrile butadiene styrene, or

polypropylene (PP). According to ASTM D2240 or ISO 868 standard, the Shore durometers of the abovementioned materials all range from 50 to 90. In this embodiment, the Shore durometer of the base body **401** ranges from 70 to 80.

[0052] The sensor assembly **5** is removable from and positioned relative to the patch base **4**. The sensor assembly **5** includes a sensor seat **501** and the sensor **502**. The sensor **502** is connected to the sensor seat **501** and is disposed in the needle **120**. The sensor seat **501** has two securing portions **503** that are respectively engaged with the link portions **781** and that are recesses (Due to the viewing angle causing components to overlap, only one securing portion **503** and one link portion **781** are shown in FIG. 3.). The support portions **302** cooperatively support the sensor seat **501** to position the sensor seat **501** relative to the main body member **51**. In another embodiment, the securing portions **503** may be protrusions, the link portions **781** may be recesses that are respectively coupled to the securing portions **503** for preventing the sensor seat **501** from rotating before the sensor **502** is inserted into the user's skin.

[0053] The peel-off component **6** is connected to the patch base **4** and the bottom cover **40**. Referring to FIGS. **11** to **13**, the peel-off component **6** is connected to the bottom plate portion **41**, and is disposed at an inner side of and is adjacent to the second notch **408** (i.e., the peel-off component **6** is disposed at an inner side of the one of the first sides **406**). Specifically, the peel-off component **6** is connected to the outer release layer **403**. The peel-off component **6** may be a type of foam tape, or another material with a specific thickness and resilience, and is adhereable on two sides. The peel-off component **6** may be a biocompatible material. In this embodiment, the peel-off component **6** is a foam tape, and the outer release layer **403** has a plurality of small holes **412**. When the bottom cover **40** is separated from the outer shell **10**, via the connection between the peel-off component **6** and the outer release layer **403**, the peel-off component **6** removes the outer release layer **403** from the adhesive pad **402**.

[0054] Referring to FIG. **12**, the patch base **4** further includes an inner release layer **403'**. The inner release layer **403'** is attached to the bonding layer **420**. The outer release layer **403** is attached detachably to the adhesive pad **402**. Referring further to FIG. **11**, the base body **401**, the bonding layer **420**, the inner release layer **403'**, the adhesive pad **402** and the outer release layer **403** are assembled through a first attaching step, a pre-hot pressing step, a second attaching step, and a hot pressing step, details of which are as follows. [0055] (1) In the first attaching step, the bonding layer **420** is attached to the patch base **4**; specifically, the inner release layer **403'** is attached to the bonding layer **420**, and the bonding layer **420** is then attached to the base body **401**. The bonding layer **420** is made of a polymeric material such as thermoplastic polyurethane (TPU), or ethylene vinyl acetate (EVA), and may be biocompatible. [0056] (2) In the pre-hot pressing step, the bonding layer **420** is processed. The pre-hot pressing step is performed from the side of the inner release layer **403'** opposite to the bonding layer **420** and is performed in a direction toward the base body **401** to secure the bonding layer **420** on a bottom surface of the base body **401**. The bonding layer **420** has at least one pre-hot pressing position **420'**. Referring to FIG. **12**, in this embodiment, the bonding layer **420** has four pre-hot pressing portions **420'**. The pre-hot pressing step is performed for 3 seconds to 10 seconds at a temperature of 75° C. to 85° C. and under a pressure of 3.5 kg/cm.sup.2 to 4.5 kg/cm.sup.2. After the pre-hot pressing step, the inner release layer **403'** is removed from the bonding layer **420** before the second attaching step is performed. [0057] (3) In the second attaching step, the adhesive pad **402** is attached to the bonding layer **420**; specifically, the outer release layer **403** is attached to the adhesive pad **402**, and the adhesive pad **402** is then attached to the bonding layer **420**. The adhesive pad **402** has an adhesive surface **402'** attached to the outer release layer **403**, and at least one hot pressing position **402''**. In this embodiment, the adhesive pad **402** has four hot pressing positions **402''**. [0058] (4) In the hot pressing step, the adhesive pad **402** is processed. The hot pressing step is performed from a side of the outer release layer **403** opposite to the adhesive pad **402** and is performed in a direction toward the base body **401** to secure the adhesive pad **402** on the base body **401** via the bonding layer **420**. The hot

pressing step is performed for 10 seconds to 20 seconds at a temperature of 115° C. to 125° C. and under a pressure of 3.5 kg/cm.sup.2 to 4.5 kg/cm.sup.2.

[0059] An adhering step of the peel-off component **6** is as follows. The peel-off component **6** is disposed between the bottom cover **40** and the outer release layer **403**, and corresponds in position to one of the hot pressing positions **402''** of the adhesive pad **402** and the pre-hot pressing positions **420'** of the bonding layer **420**.

[0060] To provide insight into effects caused by the cooperation of the components in the disclosure, technical means used, and expected results, further elaboration is as follows, which may help the reader have deeper and concrete understanding of the disclosure.

[0061] Referring to FIGS. **1** to **3**, **13** and **14**, when the insertion device is assembled, the bottom cover **40** is airtightly coupled to the outer shell **10**. At this time, the main body member **51** and the main body cap **52** are integrated with each other via engagement between the engagement portions **513** and the connection portions **521** (see FIG. **3**). When the insertion device is in an unfired state, there is a distance between the top cap **20** of the cover assembly **1** and the main body cap **52**, and the buckle portions **76** of the implant seat **70** respectively engage the buckle stopper portions **514** of the main body member **51** to keep the implant seat **70** at an upper position. The needle implantation limit structure (A) formed by the implant seat **70** and the main body member **51** produces a safety lock to lock a position of the implant seat **70** relative to the main body member **51**. The first resilient member **81** is compressed and stores a resilient force. The limit members **522** are respectively slotted in the limiter slots **752**, which limits movement of the limiter components **751** in a radial direction. The needle extraction limit structure (B) forms another safety lock that locks a position of the needle extraction member **90** relative to the implant seat **70**. The second resilient member **82** is compressed and stores a resilient force. The needle extraction member **90** is at an unfired position (see FIGS. **13** and **14**). The securing portions **503** of the sensor assembly **5** are respectively coupled to the link portions **781**, so that the sensor seat **501** is connected to the implant seat **70**. The needle **120** of the needle implant member **100** is disposed within the main body member **51**, and a tip of the needle **120** does not protrude from the patch base **4**. The patch base **4** is positioned relative to the main body member **51**. At this time, the bottom cover **40** abuts against the abutment portions **301**, which positions the fixture members **3** relative to the main body member **51**. The support portions **302** cooperatively support the sensor seat **501** to position the sensor seat **501** relative to the main body member **51**. The first hooks **303** respectively engage the engagement grooves **405**.

[0062] When the sensor **502** is to be implanted under the user's skin, an implant process is performed as follows.

[0063] Referring to FIG. **15**, the user first removes the bottom cover **40** from the outer shell **10**. The bottom cover **40** has a force exertion portion **42** for receiving a force, and a pivot portion **43** formed opposite to the force exertion portion **42**. When the force is exerted on the force exertion portion **42**, a force-exertion torque is formed, with the pivot portion **43** and the force exertion portion **42** acting collectively as a single torque arm to rotate about the pivot portion **43**. The length of the torque arm is defined by a distance between the force exertion portion **42** and the pivot portion **43**. The force-exertion torque allows the user to form a partial side opening between the bottom cover **40** and the outer shell **10** via the force exertion portion **42**. When the bottom cover **40** is being removed from the outer shell **10**, the outer release layer **403** is detached from the adhesive pad **402** via the connection between the peel-off component **6**, the bottom plate portion **41** of the bottom cover **40**, and the outer release layer **403**. After the outer release layer **403** is detached from the adhesive pad **402**, the user can adhere the adhesive pad **402** of the bottom cover **40** to a desired body part (not shown) where the sensor **502** is to be implanted.

[0064] Referring to FIGS. **16** and **17**, when the user pushes the outer shell **10** of the cover assembly **1** in a needle implantation direction (F), the outer shell **10** drives the top cap **20** and the inner lining member **30** to move towards the desired body part. Specifically, the inner lining member **30** is

moved to a lower critical position in preparation for unlocking the needle implantation limit structure (A). At this time, the main body cap **52** is not displaced.

[0065] Referring to FIGS. **18** and **19**, when the user further pushes the outer shell **10**, the push members **35** of the inner lining member **30** are driven to respectively push against the driver portions **304**, thereby driving the fixture members **3** to respectively slide outwards along the slide grooves **515**. The abovementioned movement of the components form an unlocking step of the fixture members **3** before the needle **120** is fired, which removes restrictions on the relative positions of the sensor assembly **5** and the patch base **4**.

[0066] Referring to FIGS. **20** and **21**, the outer shell **10** is further pushed so that a distance between the inner lining member **30** and the main body cap **52** is shortened or closed. At this time, the firing portions **33** of the inner lining member **30** respectively drive the buckle portions **76** of the implant seat **70** to move to be respectively disengaged from the buckle stopper portions **514** (see FIG. **21**). Meanwhile, the latch portions **34** respectively engage the lock portions **516** (see FIG. **20**), so that the inner lining member **30** is secured on the main body member **51**. The needle implantation limit structure (A) is destroyed when the buckle portions **76** of the implant seat **70** are disengaged from the buckle stopper portions **514** of the main body member **51**, which removes the restriction on the relative positions of the main body member **51** and the implant seat **70**. When the cover assembly **1** is pushed, the push members **35** of the inner lining member **30** respectively push against the driver portions **304** of the fixture members **3**, thereby driving the fixture members **3** to simultaneously slide outwards relative to the main body member **51**, so that the sensor seat **501** loses the support of the support portions **302**, and that the first hooks **303** are disengaged from the respective engagement grooves **405**. . . . The driver surfaces **305** allow smooth movement of the push members **35** in driving the fixture members **3** to move.

[0067] Referring to FIGS. **22** and **23**, when the buckle portions **76** of the implant seat **70** are disengaged from the buckle stopper portions **514** of the main body member **51**, the restriction on the relative positions of the main body member **51** and the implant seat **70** is removed so that the resilient force stored in the first resilient member **81** can be released. The resilient force stored in the first resilient member **81** biases the implant seat **70** to move away from the main body cap **52** in the needle implantation direction (F). Specifically, the implant seat **70** is biased to a lower position, the sensor assembly **5** is driven by the implant seat **70** and the needle implant member **100** to move to a fired position, where the needle **120** penetrates the skin, the sensor **502** that is carried in the needle **120** is implanted under the skin, and the sensor seat **501** of the sensor assembly **5** is engaged with the sensor positioning groove **400** of the patch base **4**. At this time, the limiter members **522** of the main body cap **52** are disengaged from the limiter slots **752**.

[0068] Afterwards, referring to FIGS. **24** and **25**, since the limiter members **522** of the main body cap **52** are disengaged from the limiter slots **752**, movement of the limiter components **751** in the radial direction is no longer restricted, thereby unlocking the positional restriction the limiter module **75** has on the needle extraction member **90**. The resilient force stored inside the second resilient member **82** is released, which biases the needle extraction member **90** to drive the needle implant member **100** to move in a needle extraction direction (R) opposite to the needle implantation direction (F), so as to extract the needle **120** from the skin and into the implant seat **70**. The positioner member **78** keeps pressing the sensor assembly **5** to the patch base **4**, which prevents the kinetic energy resulting from the needle **120** being extracted from causing the sensor assembly **5** to bounce back.

[0069] Referring to FIG. **26**, when the user separates and lifts the cover assembly **1** and the implant module **2** away from the skin and the patch base **4**, the patch base **4** and the sensor assembly **5** remain attached to the skin (only the cover assembly **1** and the implant module **2** are shown in FIG. **26**, the patch base **4** and the sensor assembly **5** that are attached to the skin are not shown).

[0070] Afterwards, the user reattaches the bottom cover **40** that was removed to the bottom of the outer shell **10**, and seals the bottom of the outer shell **10** with the bottom cover **40**.

[0071] Effects of the disclosure are as follows. [0072] 1. Referring to FIGS. 11 and 12, the peel-off component 6 is disposed at an inner side of the periphery 404 of the base body 401. The base body 401 is rigid, so that the base body 401 can serve as a supporting fixture during removal of the outer release layer 403, which causes forces to concentrate around the peel-off component 6 and be utilized more effectively allowing for concentrated peeling torque; accordingly, the success rate of removing the outer release layer 403 via the peel-off component 6 is at 99%. If the peel-off component 6 is provided on, for example, a soft textile component, the success rate of removing the outer release layer 403 via the peel-off component 6 will be greatly lowered. [0073] 2. The adhesive pad 402 has the first notch 407', and the outer release layer 403 has the second notch 408. The first notch 407' and the second notch 408 allow air to escape, which decreases resistance in the removal process of the outer release layer 403, and makes the removal process easier. Adding a notch or a gap at a periphery and/or a set location of the outer release layer 403 creates a starting point for the removal process of the outer release layer 403, making the removal process easier. [0074] 3. The outer release layer 403 has microstructures, for example the small holes 412 or small bumps, which decrease adhesion. Furthermore, holes or slots can weaken the structure of the outer release layer 403. When designing a removal path of the outer release layer 403, the inventor may consider the structural weakness thereof, so as to make the removal process easier. [0075] 4. The outer release layer 403 is attached to the adhesive pad 402 that has been hot-pressed. After the adhesive pad 402 has been hot-pressed, the portions of the adhesive pad 402 at the hot pressing positions 402" have adhesion less than that of other portions of the adhesive pad 402 that were not hot-pressed, which is advantageous for removal of the outer release layer 403 from the adhesive pad 402 via the peel-off component 6 when the bottom cover 40 is being removed from the outer shell 10.

[0076] In conclusion, when the bottom cover 40 is removed from the outer shell 10, the outer release layer 403 can be simultaneously removed from the adhesive pad 402, so that the insertion device can be quickly secured onto the user's skin, thereby achieving convenient and quick usage. Hence, the objective of the disclosure is achieved.

[0077] In the description above, for the purposes of explanation, numerous specific details have been set forth in order to provide a thorough understanding of the embodiment(s). It will be apparent, however, to one skilled in the art, that one or more other embodiments may be practiced without some of these specific details. It should also be appreciated that reference throughout this specification to "one embodiment," "an embodiment," an embodiment with an indication of an ordinal number and so forth means that a particular feature, structure, or characteristic may be included in the practice of the disclosure. It should be further appreciated that in the description, various features are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure and aiding in the understanding of various inventive aspects; such does not mean that every one of these features needs to be practiced with the presence of all the other features. In other words, in any described embodiment, when implementation of one or more features or specific details does not affect implementation of another one or more features or specific details, said one or more features may be singled out and practiced alone without said another one or more features or specific details. It should be further noted that one or more features or specific details from one embodiment may be practiced together with one or more features or specific details from another embodiment, where appropriate, in the practice of the disclosure.

[0078] While the disclosure has been described in connection with what is(are) considered the exemplary embodiment(s), it is understood that this disclosure is not limited to the disclosed embodiment(s) but is intended to cover various arrangements included within the spirit and scope of the broadest interpretation so as to encompass all such modifications and equivalent arrangements.

Claims

1. An insertion device adapted to be installed in a biosensor, said insertion device comprising: a cover assembly including an outer shell and a bottom cover that is removably coupled to said outer shell, said outer shell defining an accommodating space; an implant module disposed in said accommodating space and operable for implanting and extracting a needle; a patch base positioned removably relative to said implant module, and including a base body, an adhesive pad that is connected fixedly to said base body, and an outer release layer that is attached detachably to said adhesive pad; a sensor assembly connected to said base body when said needle is implanted by said implant module; and a peel-off component connected to said bottom cover and said outer release layer; wherein said peel-off component removes said outer release layer from said adhesive pad when said bottom cover is separated from said outer shell.
2. The insertion device as claimed in claim 1, wherein: said base body of said patch base is made of a material harder than a material of said adhesive pad; said base body has a periphery; and said peel-off component is disposed at an inner side of said periphery.
3. The insertion device as claimed in claim 2, wherein: said periphery of said base body has two first sides, and two second sides interconnecting said first sides; said adhesive pad has a first notch corresponding in position to one of said first sides; said outer release layer has a second notch corresponding in position to said first notch; and said peel-off component is disposed at an inner side of and is adjacent to said second notch.
4. The insertion device as claimed in claim 1, wherein: said patch base further includes a bonding layer disposed between said base body and said adhesive pad; said bonding layer is pre-hot-pressed and secured to said base body; and said adhesive pad is hot-pressed and connected to said base body via said bonding layer.
5. The insertion device as claimed in claim 4, wherein: said patch base further includes an inner release layer attached to said bonding layer; and said inner release layer is pre-hot-pressed from a side thereof opposite to said bonding layer to secure said bonding layer on said base body.
6. The insertion device as claimed in claim 5, wherein: said base body, said bonding layer, said inner release layer, said adhesive pad and said outer release layer are assembled through a first attaching step, a pre-hot pressing step, a second attaching step, and a hot pressing step; in the first attaching step, said inner release layer is attached to said bonding layer, and said bonding layer is then attached to said base body; the pre-hot pressing step is performed from said side of said inner release layer opposite to said bonding layer and is performed in a direction toward said base body to secure said bonding layer on said base body; after the pre-hot pressing step, said inner release layer is removed from said bonding layer before the second attaching step is performed; in the second attaching step, the outer release layer is attached to said adhesive pad, and said adhesive pad is then attached to said bonding layer; and the hot pressing step is performed from a side of said outer release layer opposite to said adhesive pad and is performed in a direction toward said base body to secure said adhesive pad on said base body via said bonding layer.
7. The insertion device as claimed in claim 6, wherein: said bonding layer is made of a polymeric material, and has at least one pre-hot pressing position; and said adhesive pad has at least one hot pressing position.
8. The insertion device as claimed in claim 7, wherein the pre-hot pressing step is performed for 3 seconds to 10 seconds at a temperature of 75° C. to 85° C. and under a pressure of 3.5 kg/cm.² to 4.5 kg/cm.².
9. The insertion device as claimed in claim 8, wherein the hot-pressing step is performed for 10 seconds to 20 seconds at a temperature of 115° C. to 125° C. and under a pressure of 3.5 kg/cm.² to 4.5 kg/cm.².
10. The insertion device as claimed in claim 7, wherein said peel-off component is disposed

between said bottom cover and said outer release layer, and corresponds in position to at least one of said at least one hot pressing position of said adhesive pad and said at least one pre-hot pressing position of said bonding layer.
